Harshit Nair

Robotics & Control Engineer

RESUME_1824

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Total Experience: 7 years

PROFESSIONAL SUMMARY

Robotics & Control Engineer with 7 years of experience in robotics, using ROS 2, Python, C++, MATLAB. Experienced in autonomous navigation, control loops and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Python, Robot Kinematics, SLAM, PLC, OpenCV, Gazebo, ROS 2, Motion Planning, C++, Simulink, PID Control, MATLAB

Methods: Autonomous navigation, Control loops, Sensor fusion

PROFESSIONAL EXPERIENCE

Senior Robotics & Control Engineer | HelioNex Labs | Jan 2022 - Present

- Reduced robotic path completion time by 29% through route and controller tuning
- Lowered navigation recovery events by 17% using improved obstacle handling
- Improved pick-position repeatability by 39% after calibration updates
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Robotics & Control Engineer | OrionPath Technologies | Jul 2019 - Dec 2021

- Improved pick-position repeatability by 34% after calibration updates
- Validated 674 robot simulation scenarios with recorded sensor logs

PROJECTS

Warehouse Mapping Trial | ROS 2, C++, Python, MATLAB

- Benchmarked LiDAR SLAM configurations under moving obstacles.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

Motor Control Testbed | Gazebo, OpenCV, ROS 2, Robot Kinematics

- Tuned PID loops and safety stops for repeatable positioning.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

EDUCATION

B.Tech in Mechatronics Engineering, Aarohan Institute of Engineering, 2019

Relevant Methods: Autonomous navigation, Control loops, Sensor fusion

CERTIFICATIONS

ROS 2 Navigation Workshop (2023)